2nd Order ODEs

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Second Order ODE -Standard Form

- General form: $F(x, y, y', y'') = 0 \quad (*)$
- Here x — independent variable, y — dependent variable
- Linear 2nd Order ODE: $a_0(x)\,y'' + a_1(x)\,y' + a_2(x)\,y = a_3(x)$.
- If $a_0(x) \neq 0$ for every x in the domain of y, divide the ODE by $a_0(x)$: $y'' + P(x)\,y' + Q(x)\,y = R(x)$.
- $P(x) = \dfrac{a_1(x)}{a_0(x)}, \quad Q(x) = \dfrac{a_2(x)}{a_0(x)}, \quad R(x) = \dfrac{a_3(x)}{a_0(x)}$

  $P(x)$, $Q(x)$ are coefficients of the DE
- **Standard Form** of the 2nd order ODE is

$$y'' + P(x)\,y' + Q(x)\,y = R(x). \tag{1}$$

- Otherwise, $(*)$ is called **nonlinear**.



Homogeneous and Non-homogeneous DE

- If $R(x) \equiv 0 \ \ \forall x$ considered in the standard form, then

  $$y'' + P(x)\, y' + Q(x)\, y = 0$$

  is called a **homogeneous** DE.

- If $R(x) \not\equiv 0$, (1) is called **non-homogeneous**.
  Examples:
  1. $y'' + 25y = e^{-x} \cos x$ (linear, non-homogeneous)

  2. $y'' + 25y = 0$ (linear, homogeneous)

  3. $xy'' + y' + xy = 0 \ \Rightarrow \ y'' + \frac{1}{x}y' + y = 0, \ x \neq 0$ (linear, homogeneous)

  4. $y''y + y'^2 = 0$ (nonlinear)



Solution of a Second Order ODE

- Consider the 2nd order DE $F(x, y, y', y'') = 0$ over I.
- $y = \phi(x)$ is a solution on I if:
  - ϕ is defined in I
  - ϕ is twice differentiable in I
  - the ODE becomes an identity if y, y', y'' are replaced by ϕ, ϕ', ϕ'' respectively.
- $y = \cos x$ is a solution of $y'' + y = 0$. Why?

  $y' = -\sin x, \quad y'' = -\cos x$

  $\text{LHS} = y'' + y = -\cos x + \cos x = 0 = \text{RHS}$

- Similarly, $y = \sin x$ is a solution of $y'' + y = 0$.
- In general, $y = C_1 y_1 + C_2 y_2$ is a solution of $y'' + y = 0$ if $y_1 = \cos x, \ \ y_2 = \sin x, \ C_1, C_2$ are arbitrary constants.



Superposition Principle

Theorem. For a homogeneous linear ODE

$$y'' + P(x)y' + Q(x)y = 0, \qquad x \in I, \tag{2}$$

any linear combination of two solutions on an open interval I is again a solution of (2) on I. (Homework)

- $V = \{y \mid y'' + P(x)y' + Q(x)y = 0\}$ is a subspace of $\mathcal{D}^2(I)$.
- $V_1 = \{y \mid y'' + P(x)y' + Q(x)y = R(x)\}$ is NOT a subspace, $R(x) \neq 0$ for some x.
- For example, $y_1 = 1 + \cos x$, $y_2 = 1 + \sin x$ are solutions of $y'' + y = 1$.
- Consider $1 \cdot y_1 + 1 \cdot y_2 = 2 + \cos x + \sin x$
- Here $y = y_1 + y_2$ is **NOT** a solution of $y'' + y = 1$, since $y'' + y = (-\cos x - \sin x) + (2 + \cos x + \sin x) = 2 \neq 1$ (RHS).
- $\therefore$ Superposition of solutions is not a solution of $y'' + y = 1$.



Example: A Nonlinear ODE

- Consider a nonlinear ODE $y''y - xy' = 0$.
- Since $(x^2)''x^2 - x(x^2)' = 2x^2 - 2x^2 = 0$, thus $y_1 = x^2$ is a solution of the DE.
- Obviously, $y_2 = 1$ satisfies the DE.
- But their sum $y_1 + y_2 = x^2 + 1$ is **not** a solution as

$$(x^2 + 1)''(x^2 + 1) - x(x^2 + 1)' = 2(x^2 + 1) - x(2x) = 2 \neq 0$$

- A constant multiple of x^2, e.g. $-x^2$, is **not** a solution of $y''y - xy' = 0$. (Homework)



Initial Value Problem (IVP)

- Consider a 2nd order linear homogeneous DE:

$$y'' + P(x)y' + Q(x)y = 0, \quad x \in I. \tag{3}$$

- Consider the initial conditions

$$y(x_0) = y_0; \quad y'(x_0) = y_0' \quad (4), \quad x_0 \in I. \tag{4}$$

- Since (3) is a 2nd order linear DE, we expect two arbitrary constants in its general solution.
- Let it general solution be $y = C_1y_1 + C_2y_2$ of (3), where y_1, y_2 are suitable solutions of (3).
- Using (4), we can obtain the two arbitrary constants C_1, C_2 to get the unique (particular) solution of the IVP.
- Here $y = y(x)$, the unique solution, passes through (x_0, y_0) and has y_0' as the slope of the tangent at (x_0, y_0).



Solution of 2nd order DEs

- **Linear Independence.** Two functions y_1, y_2 on I are **linearly independent** if

  $c_1 y_1(x) + c_2 y_2(x) = 0 \quad \forall x \in I \;\Rightarrow\; c_1 = c_2 = 0.$
- Otherwise, y_1, y_2 are **linearly dependent**.
- Equivalently, y_1, y_2 are dependent iff one is a constant multiple of the other.
- Example: e^x, e^{2x} are linearly independent on any interval.
- Example: $\sin x, 2\sin x$ are linearly dependent.
- Linear independence of solutions is key to forming the general solution.



Solution 2nd order DEs

- **Definition (Wronskian).** For functions $y_1(x), y_2(x)$ differentiable on I,
$$W(y_1, y_2)(x) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = y_1 y_2' - y_2 y_1'.$$
- W is itself a function of x on I.
- The Wronskian provides a computable test for linear independence.
- If $W(y_1, y_2)(x_0) \neq 0$ for some $x_0 \in I$, then y_1, y_2 are linearly independent on I.
- If y_1, y_2 are linearly dependent, then $W(y_1, y_2)(x) \equiv 0$ on I.
- Note: $W \equiv 0$ does *not* always imply dependence for arbitrary functions.



Solution 2nd order DEs

Theorem (Wronskian Test for Solutions). Let y_1, y_2 be solutions of $y'' + P(x)y' + Q(x)y = 0$ on I. Then y_1, y_2 are linearly independent on I if and only if $W(y_1, y_2)(x) \neq 0$ for *every* $x \in I$.

Proof: **($\Rightarrow$)** Suppose that y_1 and y_2 are linearly independent on I. Let $W(x) = y_1y_2' - y_2y_1'$ be their Wronskian. Differentiating $W(x)$, we obtain

$$\begin{aligned} W'(x) &= y_1'y_2' + y_1y_2'' - y_2'y_1' - y_2y_1'' \\ &= y_1y_2'' - y_2y_1''. \end{aligned}$$

Since both y_1 and y_2 satisfy $y'' = -P(x)y' - Q(x)y$, it follows that

$$\begin{aligned} W' &= y_1(-Py_2' - Qy_2) - y_2(-Py_1' - Qy_1) \\ &= -P(y_1y_2' - y_2y_1') \\ &= -PW. \end{aligned}$$



Proof : CTN...

Hence, $\boxed{W' + PW = 0}$ (It is known as Abel's Formula).
This is a first-order linear differential equation whose solution is $W(x) = Ce^{-\int P(x)\,dx}$, where C is a constant.
Suppose, contrary to the claim, that $W(x_0) = 0$ for some $x_0 \in I$.
Then

$$C = 0, \quad (\because e^{-\int P(x)\,dx} \neq 0)$$

and consequently $W(x) \equiv 0$ throughout I.
Assume that $y_1 \not\equiv 0$. On any interval where $y_1 \neq 0$,

$$\left(\frac{y_2}{y_1}\right)' = \frac{y_1 y_2' - y_2 y_1'}{y_1^2} = \frac{W(x)}{y_1^2} = 0.$$

Therefore, $\frac{y_2}{y_1} = k$, where k is a constant. Hence, $y_2 = ky_1$, which contradicts the assumption that y_1 and y_2 are linearly independent.
Thus, $W(x) \neq 0$ for every $x \in I$.



Proof : CTN...

($\Leftarrow$) Conversely, suppose that $W(x) \neq 0$ for every $x \in I$. Assume, to the contrary, that y_1 and y_2 are linearly dependent. Then there exist constants c_1, c_2, not both zero, such that

$$c_1 y_1 + c_2 y_2 = 0. \quad \text{non-trivial linear combination}$$

Without loss of generality, let $c_2 \neq 0$. Then $y_2 = -\frac{c_1}{c_2} y_1$ and $y_2' = -\frac{c_1}{c_2} y_1'$. Therefore,

$$\begin{aligned} W(y_1, y_2) &= y_1 y_2' - y_2 y_1' \\ &= y_1 \left(-\frac{c_1}{c_2} y_1' \right) - \left(-\frac{c_1}{c_2} y_1 \right) y_1' \\ &= 0. \end{aligned}$$

Thus,

$$W(x) \equiv 0,$$

which contradicts the assumption that $W(x) \neq 0$ for every $x \in I$. Hence, y_1 and y_2 are linearly independent.

